An Ultra-Low Frequency Noise Laser Based on All-Fiber Integrated Recirculating Interferometers

Jing Gao, Dongdong Jiao, Linbo Zhang, Guanjun Xu, Bimu Yao, Ruifang Dong, Tao Liu, and Shougang Zhang

***Abstract*—We present a pioneering method for achieving an ultra-low frequency noise laser by employing an all-fiber integrated recirculating interferometer. This innovative approach offers dual benefits: it retains the compactness of a short fiber-delay line while significantly enhancing frequency-discriminator sensitivity by over an order of magnitude, without the need to extend the fiber-delay line. Leveraging the principles of the Nth-order beat note spectrum from recirculating interferometers, we develop a multi-parameter model to optimize the core parameters, thereby achieving an interferometer with heightened frequency-discriminator sensitivity. Our experimental results are compelling: the frequency-noise power spectral density of the fiber laser is effectively reduced to below 1 Hz^2/Hz across a range of Fourier frequencies from 1 Hz to 10 kHz, utilizing a 100-meter fiber-delay line. These findings underscore the method's efficacy in reducing noise. The ultra-low frequency noise laser, as demonstrated, is not only feasible but also exhibits exceptional performance, positioning it as a strong candidate for portable applications and, notably, for space-based systems.**

***Index Terms*—High frequency-discriminator, recirculating interferometer, short fiber-delay line, ultra-low frequency noise laser.**

I. Introduction

ULTRA-LOW frequency noise lasers play an indispensable role in many applications such as quantum state manipulation [1], [2], optical clock [3], gravitational wave detection [4], [5], high-precision spectral measurement [6], dark matter detection [7], coherent communications [8], and optical fiber sensors [9]. In these applications, the laser frequency noise as a crucial parameter can affect system performance including sensitivity and accuracy [10], [11]. With the rapid development of space-based scientific engineering projects [12], [13], [14], [15], [16], such as space gravitational wave detection (such as China's "Tianqin" and "Taiji" plan, European LISA), space optical clock (SOC), and earth gravity field recovery and climate experiment (GRACE), it is of great significance to research ultra-low frequency-noise lasers with low size, weight, and power (SWaP) as sources for these missions [17].

The ultra-low frequency noise laser can be realized by providing fast frequency-noise reduction based on a fiber interferometer (FI) as a frequency-discriminator to detect the frequency fluctuations [18]. Compared with other methods of frequency-noise reduction (such as ultra-stable optical cavities [19], [20], [21], [22], [23], [24]), the FIs are lightweight, mechanically robust, simple in design, and easily compatible with other fiber systems. Therefore, the ultra-low frequency noise laser based on FIs exhibits strong competitiveness for portable applications [25], [26], [27], [28], [29], [30]. Based on an FI with the 5-km fiber-delay line, the state-of-the-art ultra-low frequency noise laser in the laboratory has achieved a frequency-noise power spectral density (PSD) less than 1 Hz^2/Hz ranging from about 0.1 Hz to 10 kHz [31]. However, the method of frequency-noise reduction typically requires a kilometer-long fiber-delay line for sufficient frequency-discriminator sensitivity of FI, mainly determined by the length of the fiber-delay line [32]. Environmental disturbances can significantly deteriorate the laser frequency-noise reduction based on the FI due to the length fluctuations of fiber-delay lines, especially at low Fourier frequencies. Theoretically, the longer the fiber-delay line, the more sensitive to environmental disturbances like air-index fluctuations, vibration, and temperature, requiring a high vacuum cavity, multiple layers of thermal shields, and relatively larger power for active temperature control [33]. For portable ultra-low frequency noise lasers, FIs with short fiber-delay lines offer the advantage of lower SWaP. However, these FIs with short fiber-delay lines lack sufficient sensitivity in frequency discrimination, which significantly impacts the performance of portable ultra-low frequency noise lasers. To enhance the performance of such lasers, it is crucial to focus on developing an FI with a short fiber-delay line and high discriminator sensitivity.

In this work, we propose a method for achieving an ultra-low frequency noise laser using all-fiber integrated recirculating interferometers (RIs) with both a short fiber-delay line and high

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frequency-discriminator sensitivity, specifically an RI with a 100 m fiber-delay line. Based on the fundamental principle of RIs, the spectrum of the Nth-order beat note is characterized and analyzed. The large frequency-discriminator quality of RIs is mainly determined by the high recirculating rate of the fiber-delay line. According to our multi-parameter optimization model with the goal of maximizing the recirculating rate, the coupling coefficient α of the coupler, and the loop optical gain γ as the RI's core parameters are optimized. Based on the optimal parameters, we construct and achieve the all-fiber and 100 m-imbalance RI with a high recirculating rate, increasing the frequency-discriminator sensitivity by more than 1 order of magnitude. Consequently, we develop an ultra-low frequency noise laser with the frequency noise PSD below 1 Hz^2/Hz from 1 Hz to 10 kHz. The experimental results demonstrate that this method can suppress the laser frequency noise and thereby achieve an ultra-low frequency noise laser. The superior performance of the proposed method makes it a promising option for portable applications of all-fiber and ultra-low frequency noise laser systems, particularly for the reduction of SWaP, which is crucial to space-based systems.

The paper is organized as follows: In Section II, a fundamental principle of a RI is presented, and the theoretical analytical model for the spectrum of the Nth-order beat note is established. In Section III, we present the experimental principle and setup in detail. Based on the theoretical analysis optimization model, the core parameters are optimized in Section IV. The experimental results and discussion of the constructed ultra-low frequency noise laser are presented in Section V. Finally, in Section VI, we present our conclusions.

II. Fundamental Principle

Since the equivalent fiber-delay length of the RI is proportional to the order number N of the beat note due to the recirculating loop, it is necessary to obtain a higher-order beat note to improve the frequency-discriminator sensitivity of the RI. In the loss-compensated RI, the spectrum of the Nth-order beat note $S_N(f)$ can be given by [34], [35]

$$S_N(f) = \frac{\gamma^N}{(1-\alpha)^2}\left[\alpha + \frac{(1-\alpha)\left(\gamma^2-\alpha\right)}{1+\gamma^2-2\gamma\cos\left[2\pi\left(f+N\Omega\right)\tau_d\right]}\right] \times S_0\left(f, N\tau_d, N\Omega\right) \quad (1)$$

where α is the coupling coefficient of the coupler, γ is the loop optical gain, Ω is the frequency shift generated by the combination of AOMs, N represents the circulation times of signal beam in the recirculating loop, namely the order number of beat note, τ_d is the delay time induced by fiber-delay line, $S_0(f, N\tau_d, N\Omega)$ is a spectrum of a conventional FI with $N\Omega$ MHz frequency shift and $N\tau_d$ delay time.

Equation (1) shows $S_N(f)$ is a modified version of $S_0(f, N\tau_d, N\Omega)$. When the α and γ are determined, the first term in the bracket of (1) is constant and the second term in the bracket of (1) is a periodical function containing a cosine function. Therefore, the $S_N(f)$ is modified by the periodical function, arising from the multi-interferences of the recirculating beams in the fiber loop. Interestingly, when the α is equal to γ^2, the second term in the bracket of (1) is zero and the $S_N(f)$ has exactly the same line shape as the beat-note spectrum from conventional FI. Thus, the parameters α and γ should be optimized to permit a precise measurement of $S_N(f)$ for the frequency-noise detection of the laser.

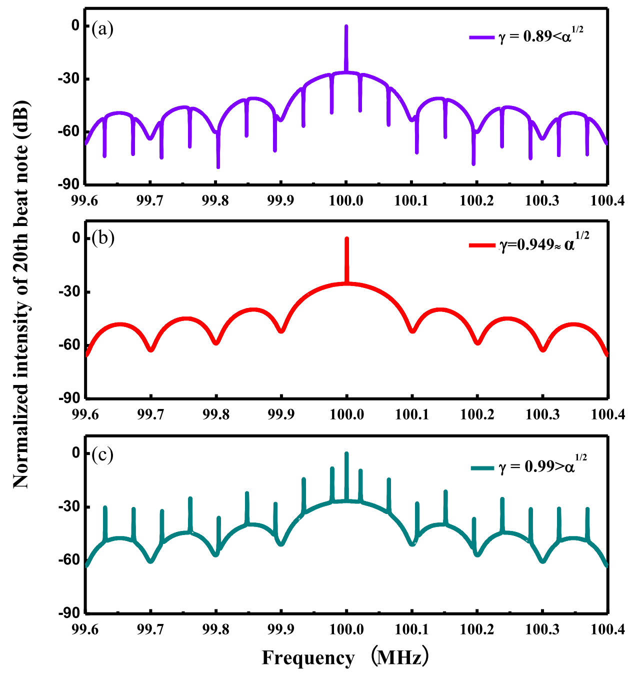

Fig. 1. The spectra of the 20th-order beat notes $S_{20}(f)$ with different values of γ when $\alpha = 0.9$, $\Omega = 5$ MHz, and $\tau_d = 0.5\ \mu$s. (a) $\gamma = 0.89 < \alpha^{1/2}$, (b) $\gamma = 0.949 \approx \alpha^{1/2}$, (c) $\gamma = 0.99 > \alpha^{1/2}$.

To clearly characterize the $S_N(f)$, we take the 20th-order beat note as an example and divide it into three categories including $\gamma < \alpha^{1/2}$, $\gamma = \alpha^{1/2}$, and $\gamma > \alpha^{1/2}$ to show the spectra of the 20th-order beat note $S_{20}(f)$. The commercially available X-type couplers provide a wide selection freedom for α, ranging from 0 to 1. Fig. 1 shows three spectra of $S_{20}(f)$ with different values of γ. As shown in Fig. 1(b), the $S_{20}(f)$ has a Lorentzian profile with a periodic coherent envelope on the wing, in which the period of the coherent envelope depends on the fiber-delay length. This line shape of $S_{20}(f)$ is similar to the beat-note spectrum of a conventional FI for $\gamma = \alpha^{1/2}$. The Fig. 1(a) and (b) show periodical peaks or notches are overlapped on the $S_{20}(f)$ due to the multi-interferences effect, in which $\gamma < \alpha^{1/2}$ and $\gamma > \alpha^{1/2}$. Generally, the random intensity fluctuations of the peaks or notches cause by the multi-interferences effect between the recirculating beams, resulting in the jitter and deformed spectra of beat notes. To measure the accuracy of the laser's frequency noise as much as possible, the stable spectrum of the beat note without peaks and notches is an optimal choice, that is, the α is equal to γ^2.

III. Experimental Setup

The schematic diagram of an ultra-low frequency noise fiber laser based on an all-fiber and 100 m imbalance-arm RI is shown

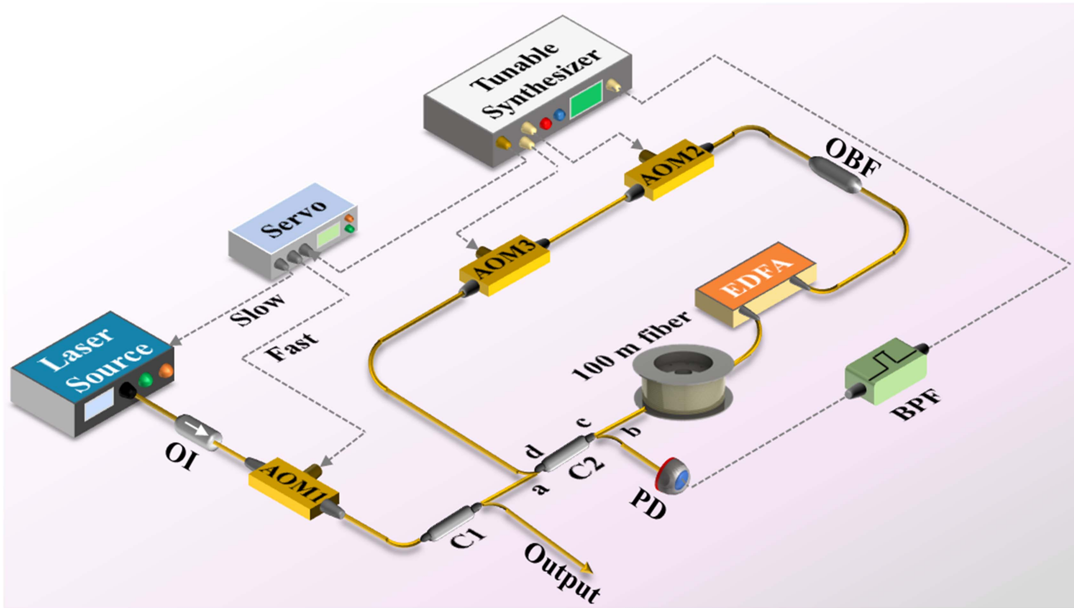

Fig. 2. Schematic diagram of an ultra-low frequency noise laser based on an RI.

in Fig. 2. The laser source is a 1550 nm erbium-doped fiber laser (NKT Photonics) with the piezoelectric transducer (PZT) modulation.

The laser beam is divided into two paths by a Y-type fiber coupler (C1) after passing an isolator (OI) and an acousto-optic modulator (AOM1). One path is entered into another fiber-optic system for the evaluation of laser performance, while the other is coupled into an RI to detect the laser frequency noise. This input beam enters into the RI from the port a of X-type C2 and is further split into two parts by the 90/10 C2. The 90% of the beam serving as the local reference beam is sent directly to a photo-detector (PD) from the port b of C2. The 10% beam acting as the signal beam is coupled to a loss-compensated fiber loop via the c port of C2 and recirculated in this loop, consisting of C2, a 100 m delay-fiber line, an erbium-doped fiber amplifier (EDFA), an optical bandpass filter (OBF), and the combined of AOMs (AOM2 and AOM3). The transmission loss of the fiber loop is compensated by an EDFA,

One concern regarding the integration of EDFA into the optical fiber loop is the potential introduction of additional phase noise with each signal pass. However, extensive research [36], [37], [38], [39], [40], [41] has demonstrated that this is not the case. The incorporation of EDFA primarily increases white noise (amplified spontaneous emission noise: ASE noise), without contributing additional phase noise [36], [37], [38], [39]. Excluding the nonlinear Kerr effect resulting from power fluctuations in the fiber delay loop, this additive white noise merely elevates the background noise level and raises the noise floor of the signal being measured [38], [39]. In our experiments, we employ an OBF to eliminate most of the ASE noise [39]. Notably, the EDFA utilized in this study is self-developed and exhibits a noise figure of approximately 3.5 dB.

The AOM1 operating at 110 MHz is utilized for laser frequency shift and correction. The AOM2 working at +50 MHz upshift and the AOM3 working at −45 MHz induce a 5 MHz frequency shift of circulating beam per pass. The radio-frequency (RF) signals that drive AOMs all come from a tunable synthesizer, which also provides a demodulation signal for the servo loop. For every circulation, the delay signal beam has a frequency shift of 5 MHz and a fiber-delay length of 100 m due to the combination of AOMs and the 100 m fiber-delay line, respectively. Meanwhile, a small portion of the delayed beam departs the delay loop via the b port of C2 and then recombines with the local reference beam. After N times circulations, the delay signal beam receives a frequency shift of 5 N MHz and a fiber-delay length of 100 Nm compared to the local reference beam. Consequently, the beat signal deriving from a heterodyne beat between the local beam and the delay beams can be detected by the PD. This beat signal consisting of a series of beat notes with frequencies of 5 N MHz and 100 Nm fiber-delay length is displayed separately in the frequency domain. A higher-order beat note exhibits a higher frequency-discriminator sensitivity due to its longer equivalent fiber-delay time or length. Consequently, the frequency-discriminator sensitivity of RI is proportional to the beat note's order. It follows that the frequency discrimination of the Nth-order beat note for the RI is N times better than the conventional FI with the same length of the fiber-delay line. By passing through an appropriate bandpass filter (BPF), a higher-order beat note can be extracted from the beat signal. This higher-order beat note is then demodulated by the tunable synthesizer and converted into a frequency correction signal through the proportional integration (PI) servo circuit. Finally, the frequency correction signal is fed back to the modulation ports of AOM1 and PZT for fast and slow frequency correction, respectively. Ultimately the frequency noise suppression is realized.

The length of the fiber-delay line is sensitive to environmental disturbances, so the detected noise from a FI is influenced by environmental coupled noise [42]. To enhance environmental noise isolation in this work, the measures are implemented. The 100 m fiber-delay line is set in an air-sealed cylindroid can. Sound-absorbing cotton, foam plastic, and sound insulation felt are utilized to absorb and dampen noise. A sealed aluminum box is deployed to effectively shield against acoustic interference and to minimize the effects of airflow jitter. To mitigate the impact of vibrations on ultra-low frequency noise lasers, this study incorporates the use of a low vibration sensitivity optical fiber spool. Finite element numerical analysis reveals that the acceleration sensitivity of the optical fiber spool in both the axial and radial directions is on the order of 10^{-11}/g. Passive temperature isolation is achieved by using a combination of an aluminum box and insulation foam, which helps to maintain a stable temperature within the system.

In this study, to reduce the thermal impact of the EDFA. The heat treatment measures implemented for the EDFA include: applying thermally conductive silica gel to the joint surface between the EDFA and the thermal conductive plate and positioning the EDFA on the box floor to enhance heat dissipation. In addition, thermal isolation is achieved between EDFA and other equipment. In the future, to further reduce the thermal impact of the EDFA, we propose several measures: substituting thermally conductive silicone rubber with thermally conductive silicone grease, replacing the bottom plate material of the box with a vapor chamber [43], establishing thermal isolation between the EDFA and other equipment, utilizing EDFA with lower power consumption, employing ultra-low expansion (ULE) glass for the fiber spool material [44], and conducting thermal design and optimization of the entire system using finite element software.

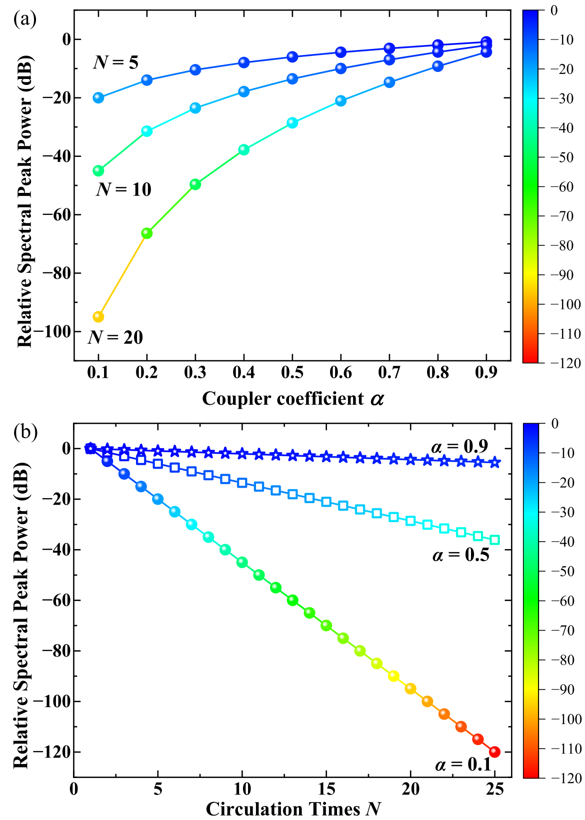

Fig. 3. The relative spectral peak intensity of beat notes as a function of the coupler coefficient α and the circulation times N. (a) The coupler coefficient α, (b) the circulation times N.

IV. Optimization of the Beat Signal

Although the $\alpha = \gamma^2$ is satisfied to avoid the deformed spectra, the spectral intensity of the Nth-order beat notes is still affected by the parameters α and γ according to (1), and then impacts the signal-to-noise ratio (SNR) of the beat note in a FI-stabilized laser system [34]. Thus, the parameters α and γ should be optimized to improve the SNR of the high-order beat notes from RI.

Based on (1), the relative spectral peak intensity P_N of the Nth-order beat note spectrum can be expressed as

$$\begin{aligned} P_N &= \frac{S_N(f)}{S_0\left(f, N\tau_d, N\Omega\right)} \\ &= \frac{\gamma^N}{(1-\alpha)^2}\left[\alpha + \frac{(1-\alpha)\left(\gamma^2-\alpha\right)}{1+\gamma^2-2\gamma\cos\left[2\pi\left(f+N\Omega\right)\tau_d\right]}\right] \end{aligned} \tag{2}$$

When the α is equal to γ^2, (2) can be expressed as follows

$$P_N(\alpha) = \frac{S_N(f)}{S_0\left(f, N\tau_d, N\Omega\right)} = \frac{\alpha^{\left(1+\frac{N}{2}\right)}}{(1-\alpha)^2} \tag{3}$$

To facilitate the embodiment of the relative spectral peak intensity, we normalize the $P_N(\alpha)$ and logarithmically process it, which can be written as

$$P_{N-lg} = 5\left(N-1\right)lg(\alpha) \tag{4}$$

Equation (4) indicates the P_N of beat notes as a function of the α and the circulation times N of the signal beam in the recirculating loop. According to (4), Fig. 3 shows that the P_N of beat notes varies with the α and N.

As depicted in Fig. 3(a), the relative spectral peak intensity P_N of beat notes exhibits an increase with the rise in the α. Moreover, it is observed that the relative spectral peak intensity of beat notes increases at a faster rate when the α is smaller. When the circulation times $N = 10$, the relative spectral peak intensity of 10th-order beat notes is about -45 dB for $\alpha = 0.1$, and the value is approximately -5 dB for $\alpha = 0.9$. As the α is 0.5, the relative spectral peak intensity of beat notes for the circulation times $N = 5$ is about -10 dB, which is approximately 30 dB larger than the value of the circulation times $N = 20$ and about 10 dB larger than the value of the circulation times $N = 10$. When the coupler coefficient α is approximately 0.9, the relative spectral peak intensity of beat notes including the circulation times $N = 5$, 10, and 20 shows a fundamental agreement.

In general, when the coupler coefficient α is determined, a linear relationship is observed between the relative spectral peak intensity of beat notes and the circulation times N. As shown in Fig. 3(b), the relative spectral peak intensity of beat notes decreases with an increase in the order number N. Additionally,

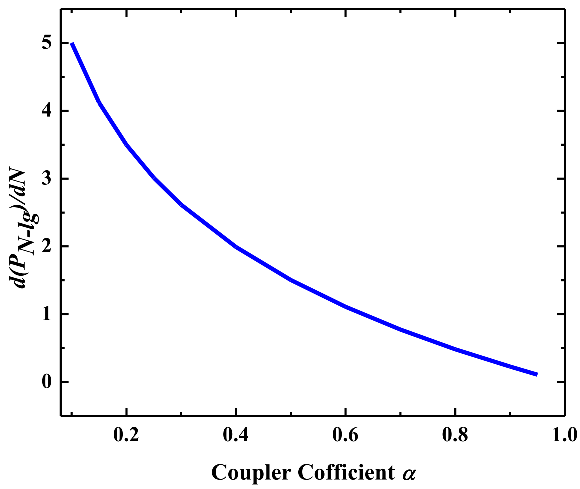

Fig. 4. The change amount of the relative spectral peak intensity of beat notes with the circulation times N as a function of the coupler coefficient α.

the attenuation trend of the relative spectral peak intensity is slowed down with a higher value of α. The relative spectral peak intensity of the 20th-order beat note is approximately 5 dB lower than that of the 1st-order beat note for $\alpha = 0.9$, whereas the relative spectral peak intensity is reduced by about 31 dB and 104 dB for $\alpha = 0.5$ and $\alpha = 0.1$, respectively. This indicates that a larger value of α corresponds to a higher circulation rate.

To achieve the beat signal with a high circulation rate and SNR, it is important to ensure a high circulation time N while maximizing the relative spectral peak intensity of beat notes. Therefore, the optimization goal is to minimize the slope of the relative spectral peak intensity of beat notes with the circulation times N. Assuming that the circulation times N is continuous, according to (4), it is easy to calculate the slope of the relative spectral peak intensity of beat notes with respect to the circulation times N as

$$\frac{d(P_{N-lg})}{dN} = 5lg(\alpha) \tag{5}$$

Based on (5), Fig. 4 displays the change amount $d(P_{N-lg})/dN$ of the relative spectral peak intensity of beat notes with the circulation times N as a function of α. When the α increases, the change amount $d(P_{N-lg})/dN$ of the relative spectral peak intensity of beat notes decreases. Conversely, when the α is smaller, the change amount $d(P_{N-lg})/dN$ decreases at a faster rate. For instance, when the coupler coefficient α is 0.2, the change amount $d(P_{N-lg})/dN$ is approximately 3.5. However, when the coupler coefficient α is 0.8, the change amount $d(P_{N-lg})/dN$ of the relative spectral peak intensity of beat notes is approximately 0.5. Finally, considering both the circulation rate, convenience of device selection, and SNR of the beat signal, the α is set to be 0.9, by which only 10% of the laser beam as the signal beam goes into the fiber loop. The value of $\gamma = 0.95 \approx \alpha^{1/2}$ can be optimized by adjusting the intensity gain of the EDFA to compensate for the transmission loss and obtain stable beat notes.

Another important parameter is the Ω generated by the combined AOMs, consisting of one up-shift AOM2 and one down-shift AOM3. These combined AOMs are inserted in the delay loop to distinguish the different beat notes with different center frequencies. Moreover, it avoids overlap between two neighboring beat notes and allows the PD to monitor as many beat signals as possible in the detection bandwidth.

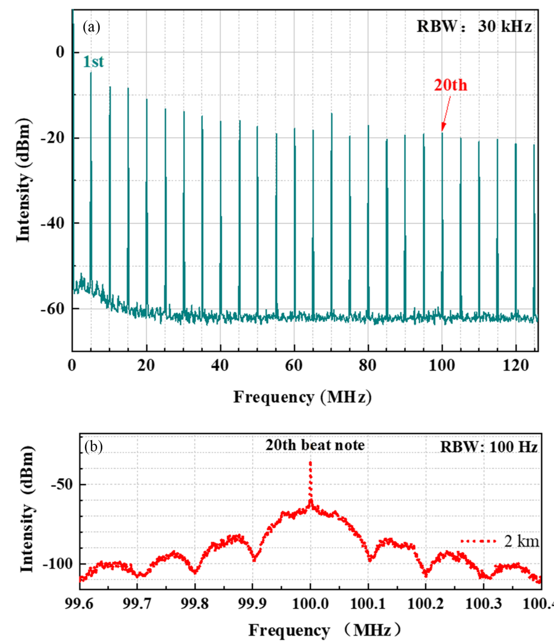

Fig. 5. The self-heterodyne beat signal of the RI. (a) The beat notes of the top twenty-fifth, (b) the spectrum of 20th-order beat note.

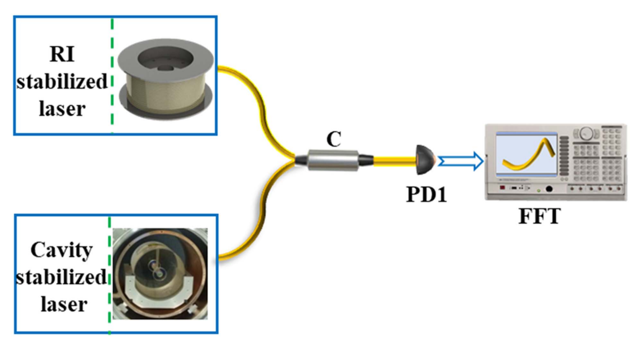

Fig. 6. The principle schematic diagram of frequency noise measurements of ultra-low frequency noise laser based on the RI. The cavity-stabilized laser denotes the laser stabilization system based on the ultra-stable optical cavity. The RI-stabilized laser represents an ultra-low frequency noise laser based on the RI. C denotes the fiber coupler.

V. Experimental Results and Discussion

A. Beat Signal

By optimizing the system parameters above, a series of stable beat notes with different fiber-delay lengths are observed simultaneously from the RI with a 100 m fiber-delay line. As shown in Fig. 5(a), 25 orders of beat notes are measured and the interval between adjacent beat notes is 5 MHz on a spectrum analyzer (FSH4) with resolution bandwidth (RBW) of 30 kHz.

The SNR of these beat notes is sufficiently high to detect the laser frequency noise. The attenuation trend of the higher-order beat notes is basically consistent with the theoretical simulation for $\alpha = 0.9$, where the SNR of the 20th-order beat note is approximately 10 dB lower than that of the 1st-order beat note. In this experiment, L represents the length of the fiber-delay line and its value is 100 m. As shown in Fig. 5, the equivalent fiber-delay length of the Nth-order beat note is N times longer than the length of the fiber-delay line due to the multipass transmission in the fiber loop. It follows that the frequency-discriminator sensitivity of the 20th-order beat note from RI has increased by 20 times, corresponding frequency-discriminator sensitivity from FI with a 2 km fiber-delay line. Since the corresponding fiber-delay lengths of the above beat notes are all far less than the coherence length of the laser source, the spectra of high-order beat notes exhibit the periodic envelopes in both wings, of which the period is inversely proportional to the fiber-delay length [45]. Fig. 5(b) shows the spectrum of the 20th-order beat note detected by improving the RBW to 100 Hz. As predicted by the theoretical simulation in Fig. 1(b), this detected spectrum of the 20th beat note is a Lorentzian line shape with the periodic envelope on the wing, in which the period of the coherent envelope is 100 kHz and the equivalent fiber-delay length is 2 km.

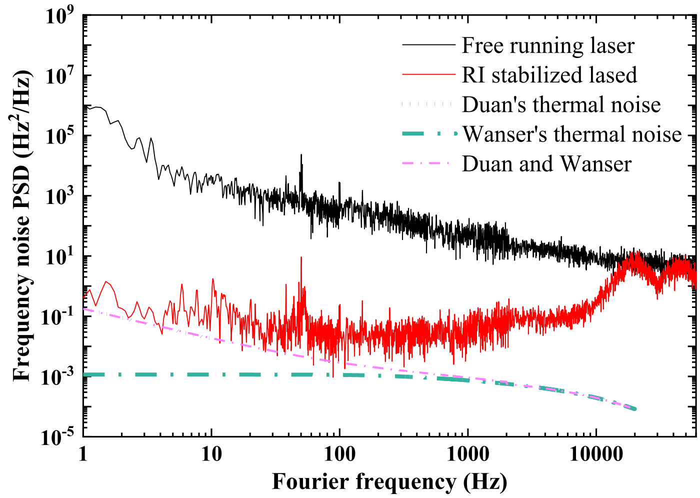

Fig. 7. The measured result of frequency noise PSD of the ultra-low frequency noise laser. The black solid line denotes the frequency noise PSD of the free running laser. The red solid line represents the frequency noise PSD of the RI-stabilized laser. The blue short dot line and the green dash-dot line are the intrinsic thermal noise effects of the RI-stabilized laser calculated by Duan's theory and Wanser's theory, respectively. The magenta dash-dot line is the thermal noise that combines Duan's theory with Wanser's theory.

B. Frequency Noise of the Ultra-Low Frequency Noise Laser

We take the 20th-order beat note of RI as an example to detect the laser's frequency noise and suppress it by the active feedback system. To evaluate the performance of the ultra-low frequency noise laser based on the RI, the frequency noise PSD of the RI-stabilized laser is measured by comparing it with a cavity-stabilized laser, constructed with a 100 mm length ultra-stable optical cavity with low drift. The fineness of the ultra-stable optical cavity is about 417000 [46]. The linewidth of the cavity-stabilized laser is approximately 0.3 Hz, and the modified Allan deviation of the fractional frequency instability is 10^{-16} orders of magnitude at 1 s–10 s averaging time [46]. The principle schematic diagram of frequency noise measurements is illustrated in Fig. 6. The beat signal of the two systems is detected by a broadband PD1 and the frequency is downconverted to about 80 kHz by mixing it with an RF signal produced by the self-developed tunable synthesizer. The contribution of the optical reference cavity stabilization system to the measurements can be considered negligible. This frequency-noise measurement method exhibits a lower noise floor especially at low Fourier frequencies (<100 Hz) than the previous two interferometers method [47].

The frequency noise PSD of the RI-stabilized laser is measured by a fast Fourier transformation (FFT: SR785) as shown in Fig. 7. We use Duan's theory and Wanser's theory to calculate the intrinsic thermal noise effect of the RI-stabilized laser [48], [49], where the corresponding fiber-delay length is 2 km.

Compared with the frequency-noise PSD of the free running laser, the frequency-noise PSD of the RI-stabilized laser is reduced by approximately 60 dB at 1 Hz. Nevertheless, the frequency noise PSD is below 1 Hz^2/Hz ranging from 1 Hz to 10 kHz, and reaches below 0.01 Hz^2/Hz at 100 Hz. The measured results indicate that the frequency noise of the RI-stabilized laser closely matches the intrinsic thermal noise calculated by Duan's theory in the frequency range of approximately 1 Hz to 10 Hz. However, there is a certain disparity when compared to the calculated results based on Wanser's theory. This difference can be attributed to the fact that Duan's theory and Wanser's theory are designed for different frequency ranges [32], [33], [50]. The thermal noise that combines Duan's theory with Wanser's theory shows a higher degree of consistency with experimental results.

This work focuses on the frequency noise PSD of the ultra-low frequency noise laser. In addition, to further evaluate the performance of the ultra-low frequency noise laser based on the RI, the modified Allan deviation of the fractional frequency instability of the RI-stabilized laser is also measured by a frequency counter (53230A) from 0.1 s to 100 s. Noting that the system is not yet fully optimized. The fractional frequency instabilities are approximately $8.5 \times 10^{-15}/0.1\,\text{s}$, $8 \times 10^{-14}/1\,\text{s}$, and $6 \times 10^{-12}/100\,\text{s}$, respectively. Currently, the primary limitation of this system is long-term frequency stability, which is predominantly affected by temperature variations. The observed temperature fluctuations likely originate from both the laboratory environment and the system itself. To enhance the long-term stability of the system in the future, we plan to implement several measures, including the use of a vapor chamber, heat source isolation, lower power components, ULE fiber spool, and thermal design optimization of the entire system.

VI. Conclusion

In summary, we propose a method for achieving an ultra-low frequency noise laser based on an all-fiber integrated RI, which has the advantage of both a short fiber-delay line and high frequency-discriminator sensitivity. The spectrum of the

Nth-order beat note is characterized and analyzed using the fundamental principle of RIs. It has been observed that the high recirculating rate plays a crucial role in determining the high frequency-discriminator quality of RIs. According to the optimal RI's core parameters including the coupling coefficient $\alpha = 0.9$ and the loop optical gain $\gamma = 0.95$, which is realized by our multi-parameter theoretical optimization model with a single objective, we develop and achieve the all-fiber high recirculating rate RI with both a 100 m fiber-delay line and high frequency-discriminator sensitivity, which is improved to over one order of magnitude while maintaining the short fiber-delay line. Consequently, we achieve ultra-low frequency noise lasers with the frequency noise PSD 1 Hz^2/Hz for Fourier frequencies ranging from 1 Hz to 10 kHz. The experimental results confirm the feasibility of the proposed method, which is not restricted by the length of the fiber-delay line. The method can suppress the laser frequency noise and achieve an ultra-low frequency noise laser. Therefore, the proposed method makes it one of the powerful candidates for portable applications, especially for space-based systems with crucial SWaP requirements.

To further reduce the system's size in the future, we will implement a series of measures, including the use of smaller-size EDFA, a ULE compact fiber spool, and advancements in optical system micro-assembly technology, and the integrated design of optical, mechanical, electrical and thermal of the whole system.

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